

Operational Limits of Ontological Innovation in LLMs: Evidence from Controlled Sensory-Modality Generation

Jarno Matarmaa^{1*}

^{1*}Institute of Radio Electronics and Information Technologies, Ural Federal University, Yekaterinburg, Russia.

Corresponding author’s email: iarnoolavi.matarmaa@urfu.ru;

Abstract

Contemporary large language models generate text that can appear conceptually novel, raising the question of whether outputs reflect genuine ontological innovation—the introduction of concepts that cannot be expressed as recombinations or transformations of training-derived primitives—or sophisticated recombination within a fixed representational manifold. We test This via a controlled experiment in an eight-modality sensory ontology. Seven language models were prompted 24 times each to propose a genuinely new ninth modality. The 168 raw response records were completion- and length-filtered, then normalized by proposal name, yielding 79 distinct proposals for analysis. Outputs were evaluated with structural decomposition, literature traceability ($\tau = \mathbf{0.40}$), embedding-space convex-hull analysis, and cosine similarity. No normalized proposal met the study’s strict novelty classification: all 79 lay inside the three-component Sentence-BERT convex hull, 48/79 showed at least one structural recombination marker, and 53/79 retrieved a scholarly-corpus match. The evidence supports a constrained-manifold interpretation for this task and analysis protocol, while not establishing a universal limit on model creativity.

Keywords: ontological innovation, computational creativity, large language models, conceptual spaces, traceability analysis, representational constraints

1 Introduction

The capacity to generate fundamentally new concepts—not merely recombinations of existing ones—has long been considered a hallmark of human intelligence and a defining test of genuine creativity. As large language models (LLMs) achieve impressive performance on a broadening range of generative tasks, a pressing question has emerged: do these systems exhibit *ontological innovation*, generating ideas that lie outside the conceptual closure of their training environment, or do they operate as extremely capable recombinators constrained to a fixed representational manifold?

This question has both philosophical and practical stakes. Philosophically, it bears on the computational theory of mind, the nature of creativity, and whether algorithmic systems can in principle exceed the representational resources with which they were trained. Practically, it determines what roles LLMs can occupy in scientific discovery, design, and other domains where genuinely new concepts—not just novel arrangements of existing ones—are required [1].

The challenge is that outputs from modern LLMs frequently *appear* novel. Proposals that introduce unfamiliar terminology, combine concepts across disciplinary boundaries, or describe mechanisms not explicitly present in any single training document can pass casual inspection as genuinely creative [2, 3]. Evaluating whether such outputs constitute true conceptual innovation requires a controlled experimental setup and rigorous multi-method analysis rather than surface-level assessment.

In this paper we report a focused empirical study designed to provide exactly this kind of evaluation. We use a multimodal sensory ontology—a well-defined, eight-modality conceptual space representing biological and extended machine sensing—as the experimental universe. This setup provides precise control: we know exactly which concepts exist in the training context, enabling systematic investigation of whether AI-generated proposals lie outside the defined space. Seven contemporary LLMs were prompted to propose a *ninth* sensory modality, genuinely beyond any of the eight. We evaluate their proposals through four convergent methods that progressively sharpen the diagnostic picture.

Our main contributions are:

1. A controlled experimental paradigm for operationalising ontological innovation, grounded in a well-defined multi-modal sensory ontology that functions as an explicit training universe.
2. Multi-method evaluation combining literature traceability with calibrated thresholds, structural decomposition, embedding-space convex-hull analysis, and continuous novelty scoring, providing convergent evidence without reliance on subjective expert judgment.
3. Systematic empirical evidence that seven state-of-the-art LLMs produce zero strictly novel proposals across 79 distinct normalized outputs, with universal hull membership and frequent structural recombination and scholarly-corpus matches.
4. Formal grounding through the *No New Basis Theorem*, relating observed empirical constraints to architectural properties of gradient-descent-trained neural networks.

The paper is organized as follows. Section 2 reviews related work on computational creativity and representational limits. Section 3 describes the dataset, task, models,

and evaluation procedures. Section 4 presents the results across all evaluation methods. Section 5 interprets the findings, and Section 6 concludes.

2 Background and Related Work

2.1 Boden’s Creativity Taxonomy

The dominant framework for analysing machine creativity is Margaret Boden’s tripartite taxonomy [4], which distinguishes outputs based on their relationship to the conceptual space from which they arise. *Combinational creativity* produces novel combinations of familiar ideas—poetic metaphors, analogies, conceptual blends. *Exploratory creativity* systematically searches well-defined conceptual spaces to find previously unvisited regions. *Transformational creativity*, the most radical category, modifies the conceptual space itself by relaxing or changing its defining constraints.

Research on AI systems has demonstrated substantial competence at combinational and exploratory creativity [5, 6]. Systems such as Copycat, JAPE, and AARON can generate outputs that human evaluators judge as genuinely surprising and apt within their respective domains. Robust transformational creativity, however, remains rare: most documented examples involve superficial parameter modifications rather than fundamental reconceptualisation [4].

Wiggins [2] introduced a formal framework for evaluating creativity claims and identified a “framing” problem: systems appear to transform their conceptual space while actually operating within an implicitly broader space defined by human designers. This critique applies with particular force to contemporary LLMs, whose training corpora implicitly define a vast but still finite conceptual universe.

2.2 Our Extension: Ontological Creativity

We argue that a fourth category, which we term *ontological creativity*, is needed to capture the most radical form of conceptual innovation. Transformational creativity modifies operators Ω or relations $\mathcal{E}$ within a conceptual space $\mathcal{C} = (V, \mathcal{E}, \Phi, \Omega)$; ontological creativity introduces new *primitive* concepts $v' \in V'$ where $V' \not\subseteq V$ and critically V' cannot be expressed as $\Omega(V)$ for any existing operators.

This distinction maps onto Ryle’s philosophical notion of category mistakes [7]: ontological creativity involves recognising that existing categories are insufficient and proposing new ones that cannot be defined by recombining or transforming current primitives. Einstein’s reconceptualisation of absolute simultaneity as a category error, leading to a revised ontology of relativistic spacetime, exemplifies this form of creativity. Generating a genuinely new sensory modality would be analogous: not an extension or hybridisation of existing ones, but a proposal for a sensing type with a physical basis and information content that cannot be expressed as a composition of the existing eight.

2.3 Formal Representational Constraints

The computational theory of mind [8, 9] and its critics [10–12] have long debated whether algorithmic systems face principled limits on what they can represent and

understand. Recent work in mechanistic interpretability provides empirical traction on this question by revealing that generation in transformer-based LLMs proceeds through retrieval and interpolation among training examples rather than through any inference mechanism that could introduce entirely new primitives [13, 14]. The transformer architecture [15] encodes all knowledge implicitly in weights learned from distributional co-occurrence, which grounds the output space in the training vocabulary of concepts. Gärdenfors’s conceptual-spaces formalism [16] provides complementary geometric intuition: a learned concept lives as a region in a quality space, and generating a new concept outside all existing regions requires an external basis vector not present in the training data.

Claim (No New Basis Theorem under fixed-basis assumptions). For a neural architecture $\mathcal{A}$ operating over representation space $\mathbb{R}^d$, with generative process $G: \mathbb{R}^d \rightarrow \mathbb{R}^d$ learned from training data $\mathcal{D} = \{x_1, \dots, x_N\}$, generated outputs $x' = G(z)$ remain within a manifold $\mathcal{M}'$ that is a continuous deformation of $\mathcal{M} = \text{span}(\{x_1, \dots, x_N\})$, such that $\dim(\mathcal{M}') \leq \dim(\mathcal{M})$.

Assumptions. (i) fixed representational basis and vocabulary during inference, (ii) gradient-based continuous learning dynamics, and (iii) no external mechanism that appends new primitive basis vectors at inference time.

Argument (informal proof sketch). First, under continuous gradient updates, learned mappings deform regions of the training-induced manifold rather than creating independent coordinate axes. Second, generative decoding composes these learned maps and therefore remains constrained by the same basis. Third, introducing ontological primitives outside the learned span would require explicit basis augmentation (new symbols, new sensors, or external memory with independent representational coordinates), which is excluded by assumptions (i)–(iii).

Consequence. Ontological novelty, defined as basis expansion rather than recombination, cannot be produced by the internal generation mechanism alone under these assumptions.

This theorem explains why even highly capable models that produce surprising combinations remain formally constrained to the conceptual universe their training data defines. Our empirical study provides a controlled test of whether this theoretical constraint manifests behaviorally.

2.4 Evaluation Challenges in Computational Creativity

A persistent methodological challenge in computational creativity research is evaluation [2, 3]. Human expert judgments are subjective, difficult to reproduce, and sensitive to framing. Recent studies show that LLMs can achieve human-level performance on divergent-thinking tasks such as the Alternative Uses Test [17], but scoring quality depends strongly on the evaluation metric [18]. Recent 2024–2026 work further confirms metric sensitivity and proposes newer creativity-evaluation protocols for LLMs, including human-comparative, temperature-sensitivity, and model-guided creativity-judging paradigms [19–25]. Our multi-method approach—traceability search, structural decomposition, embedding-space geometry, and continuous scoring—provides

Table 1 Eight-modality sensory ontology used as the experimental universe. Each modality is defined by its physical basis, representational dimensionality, and functional role.

Modality	Physical basis	Functional role	Dim.
Visual	EM spectrum 300–1000 nm (5 nm steps)	Scene and object recognition	140
Auditory	Pressure waves 1 Hz–100 kHz (128-bin FFT)	Sound source localisation	128
Tactile	Mechanical vibrations 5–800 Hz	Surface texture and pressure	32
Olfactory	Chemical categories (10 classes)	Substance identification	10
Gustatory	Taste dimensions (sweet/sour/salty/bitter/umami)	Ingestion guidance	5
Proprioceptive	20 joint angles + 10 muscle tensions	Body-state estimation	30
Vestibular	3-axis acceleration + 3-axis angular velocity	Balance and motion	6
Interoceptive	HR, respiration, skin conductance, temperature, hunger, thirst, fatigue, pain	Homeostatic regulation	8
<i>Total feature dimensionality per timestep</i>			359

Note on Dim. Dimensionality values represent the number of independently controllable encoding parameters for each modality stream (e.g., spectral bins, frequency bins, category classes, kinematic channels, and homeostatic channels), not biological receptor counts.

convergent automated evidence that can be replicated and extended. Calibrated traceability thresholds are used as empirically justified operating points for interpretation rather than as unconditional binary boundaries.

3 Materials and Methods

3.1 The Multimodal Sensory Ontology

The experimental universe consists of eight sensory modalities, each defined by a physical basis, an information channel, and a functional role in an agent’s interaction with its environment (Table 1). Modalities span biological human senses and extended machine-sensing capabilities:

The full dataset contains 5,000 training, 1,000 validation, and 1,000 test examples, each representing a 60-second episode at 1 Hz sampling (21,540 data points per example) drawn from seven everyday scenarios (walking, running, sitting, eating, drinking, listening, viewing). An explicit machine-readable ontology layer documents the physical basis, information content, and functional role of each modality, enabling automated evaluation.

3.2 Task Specification and Prompt Design

A task specification (prompt contract) is a structured set of constraints that every prompt instantiation must satisfy, ensuring comparability across models and runs.

Each model was asked to *propose a ninth sensory modality that is genuinely new and not reducible to combinations, extensions, or transformations of the existing eight.*

Proposals were required to specify: (1) the physical phenomenon to be sensed, (2) the information channel and encoding, and (3) the functional difference from existing modalities. The complete ontology table (Table 1) was provided in structured tabular format alongside a natural-language description of each modality.

Prompts were designed to be task-neutral—they did not suggest what kinds of answers would or would not qualify as novel, to avoid anchoring models toward specific response types. Twenty-four independent calls were collected from each of seven models, yielding 168 raw response records. The primary analysis unit was the unique normalized proposal name after excluding incomplete, short, and duplicate-name responses; this produced 79 distinct proposals.

3.3 Models and Implementation

Seven state-of-the-art models were evaluated: GPT-5.2 [26], Claude-Sonnet-5, Gemini-3.1-Pro-Preview [27], LLaMA-3.3-70B-Instruct [28], DeepSeek-v3.2 [29, 30], Mistral-Large [31], and Perplexity Sonar Pro, accessed via the OpenRouter, Perplexity, and Mistral APIs. All runs used temperature $T = 0.7$; the Perplexity API does not support temperature adjustment and was operated at provider defaults. System prompts were crafted to neutrally present the task without introducing bias toward particular answer types.

3.4 Evaluation Protocol

Literature traceability.

Each proposal was matched against a curated scholarly corpus of 125 documents covering sensory biology, robotics sensing, and computational neuroscience, using cosine similarity over Sentence-BERT embeddings [32]. A proposal is classified as *traceable* if its best-match similarity exceeds threshold τ .

The corpus was assembled from OpenAlex [33] and Crossref API retrieval using domain seed queries, then deduplicated and length-filtered before embedding. OpenAlex and Crossref were used as discovery/indexing layers for scholarly records, while traceability labels were assigned by semantic similarity on cleaned text rather than by keyword overlap alone.

The corpus scope is deliberately academic-mainstream. In this study, traceability is defined with respect to scientific literature in those three domains rather than science-fiction, patent, or broader philosophical corpora.

The main analysis used $\tau = 0.40$ as a calibrated retrieval threshold. It is an operational retrieval criterion, not an absolute test of ontological status: a proposal without a retrieved match can still fail the independent structural and geometric novelty checks.

Structural decomposition.

Each proposal was analysed for whether its key claims can be expressed using only the eight training primitives through: (i) range extension (preserving the sensing type while shifting its physical parameter range), (ii) hybrid combination (combining features from two or more existing modalities), or (iii) functional recombination

(repurposing an existing sensing mechanism for a different functional role). Proposals not fitting any category were labelled *structurally novel*.

Embedding-space convex-hull analysis.

Sentence-BERT embeddings were computed for each proposal and for the eight training-modality descriptions. Principal component analysis reduced both sets to three components; convex-hull membership was computed in 3D using half-space equations with numerical tolerance. A proposal classified as *outside* the hull would occupy a semantically isolated region suggesting basis expansion beyond the training ontology.

Continuous novelty scoring.

the convex-hull boundary. Proposals with $s \geq 0.5$ were classified as strictly novel under A continuous novelty proxy $s \in [0, 1]$ was computed as inverse maximum cosine similarity, discounted by the number of structural, literature-retrieval, and hull-membership constraints that were violated. The final strict classification remained rule-based: a proposal was not strictly novel when it was structurally decomposable, inside the training hull, or both.

4 Results

4.1 Primary Outcomes

The collection yielded 168 raw response records. Completion and length checks, followed by normalization of proposal names, produced 79 distinct proposals for analysis. None met the study’s strict novelty classification. This result does not mean that every proposal was literature-traceable: it follows from the combined decision rule, especially the universal embedding-hull membership described below.

At $\tau = 0.40$, 53 of 79 proposals (67.1%) retrieved at least one document from the scholarly corpus; each retrieved proposal had a strongest match in the strong-evidence tier. The mean best-match score among retrieved proposals was 0.479. Thus literature retrieval supplies supporting evidence for familiar conceptual content, rather than a saturated binary label.

4.2 Structural Decomposition

Of the 79 distinct proposals, 48 (60.8%) were structurally decomposable into training-ontology primitives. The overlapping indicators were:

- **Range extension** (24 cases, 30.4%): proposals that extend the physical parameter range of an existing modality (e.g., far-infrared thermal sensing as an extension of visual EM-band coverage, or infrasound-extended auditory detection).
- **Hybrid combination** (45 cases, 57.0%): proposals combining features from two or more existing modalities (e.g., combined tactile-proprioceptive “haptic flow” sensing, or olfactory-gustatory integration).

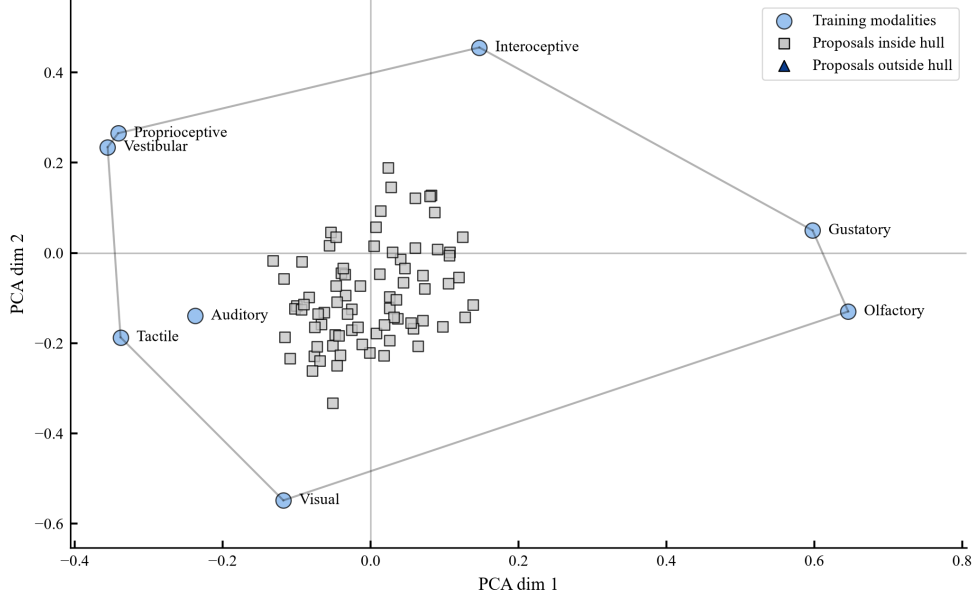


Fig. 1 Embedding-space analysis for proposals and training modalities. Convex-hull membership is computed in PCA-reduced 3D space; the 2D boundary shown is illustrative only. All 79 normalized proposal markers cluster near the interior of the training manifold—none escape to a semantically isolated region.

- **Functional recombination** (14 cases, 17.7%): proposals repurposing an existing sensing mechanism for a different functional role (e.g., using interoceptive signals for external-environment monitoring rather than homeostatic regulation).

The category counts overlap, so they should not be added. Structural decomposition alone is therefore not treated as a complete novelty test; it is interpreted jointly with embedding-space and retrieval evidence.

4.3 Embedding-Space Geometry

All 79 proposals (100%) were classified as inside the 3D convex hull of training-modality embeddings. Proposal markers cluster in the interior region of the training manifold; no proposal occupies a semantically isolated region that would suggest basis expansion beyond the training ontology.

The mean cosine similarity between each proposal and its closest training modality was $\bar{d} = 0.366$ ($SD = 0.079$), indicating substantial semantic overlap with training concepts. Figure 1 presents the embedding-space analysis with proposals visualised in the first two principal components and the 3D hull boundary shown as a visual aid.

Figure 2 provides the aggregated cosine-similarity heatmap of proposed modalities against the eight training modalities, reaching the same conclusion from a complementary metric: every proposal records elevated similarity to one or more training modalities, with no proposal showing uniformly low similarity across all eight.

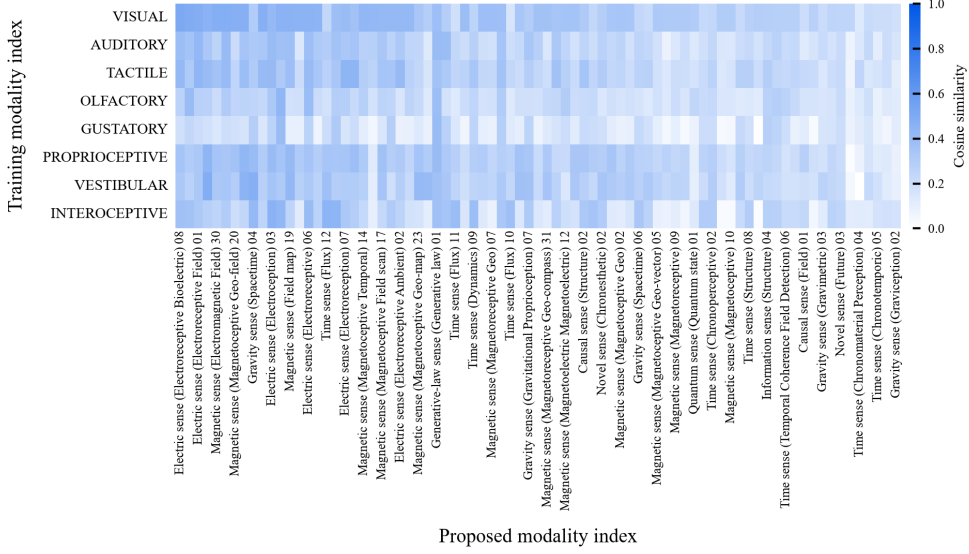


Fig. 2 Cosine similarity of proposed modalities to the eight training modalities. Columns: 79 unique normalized proposals; rows: the eight training modalities. No proposal appears uniformly dissimilar to all eight training modalities.

4.4 Cross-Model Comparison

The normalized analysis set contains different numbers of distinct proposal names per model, so the present rerun is not used for model-level inferential comparisons. Descriptively, the same strict-novelty outcome held across the pooled set, but future work should preserve a pre-specified sampling unit and use a dedicated replicated design before making cross-model effect-size claims.

4.5 Summary of Results

The evaluation methods converge on a bounded result: across 79 normalized proposals from seven LLMs, zero proposals satisfy the strict novelty classification; all lie inside the training-derived convex hull; 60.8% display at least one structural recombination marker; and 67.1% retrieve a scholarly-corpus match at $\tau = 0.40$. The evidence concerns this prompt, ontology, corpus, and analysis pipeline, and should not be read as a universal statement about model creativity.

5 Discussion

5.1 Constrained Manifold Interpretation

The consistent strict-null result across the pooled normalized proposals supports a *constrained-manifold* interpretation: under this prompt and ontology, the generated proposals remain semantically close to the training description set rather than demonstrating an independently established basis extension.

This interpretation aligns with the No New Basis Theorem stated in Section 2. The theorem motivates the observed constrained-manifold pattern under its fixed-basis assumptions. Our empirical results are consistent with this interpretation, but do not establish that any architecture is incapable of ontological innovation in every task or representation.

Gödel’s incompleteness results [9] and Turing’s halting-problem proof [8] provide deeper formal context. If recognising the inadequacy of one’s own conceptual framework requires a form of self-reference analogous to constructing a Gödel sentence, then this capacity may be unavailable to computational systems operating within fixed formal specifications. Our empirical evidence is consistent with this theoretical prediction, though it does not constitute a formal proof that such incapacity is necessary rather than contingent.

5.2 Relationship to the Creativity Taxonomy

Within Boden’s taxonomy [4], the structural decomposition results place nearly all proposals firmly in the combinational and exploratory categories: proposals are either novel combinations of existing primitives (hybrid combinations) or new regions of existing modality spaces (range extensions). A small fraction involve functional recombination, which could be viewed as approaching transformational creativity, but the transformations remain within the ontological type system defined by the training modalities rather than introducing new types.

The absence of any proposal in the ontological creativity category supports Wiggins’s [2] observation that apparent transformational creativity in AI systems frequently reflects an implicitly broader framing space defined by human designers. In our setup the framing space is made explicit through the eight-modality ontology, enabling a precise test: proposals must introduce a sensing type genuinely outside this space. None do.

5.3 Alternative Explanations

Several alternative explanations deserve consideration:

Model coverage.

All seven models span a range of commercial and open-weight architectures, but the current normalized analysis was not designed for inferential comparison between them. The pooled result should therefore not be interpreted as a scale-effect estimate.

Prompt sensitivity.

The present study used a fixed prompt design with temperature ablation limited to $T = 0.7$. We cannot rule out that alternative prompt formulations might yield marginally higher novelty scores, though they would need to change not just lexical generation patterns but the geometric structure of the output manifold. Future work should include systematic prompt-paraphrase grids to test robustness.

Evaluation sensitivity.

Conservative thresholds mean we may underestimate AI creativity by classifying some proposals as traceable that an expert would consider novel. Conversely, overly strict thresholds can collapse traceability labels and reduce interpretive value. We therefore treat thresholding as a calibrated measurement device for locating current position and movement within the response–corpus similarity landscape, not as an unconditional criterion of ontological status. However, the embedding-space analysis is threshold-free, and 100% hull membership provides threshold-independent evidence of constrained generation.

Recombination as the norm.

One counterargument holds that human ontological creativity may also, on careful analysis, involve recombination of prior concepts in ways we fail to recognise due to incomplete historical knowledge of our own cognitive processes. We cannot definitively rule this out, but the structural patterns here—range extension, hybrid combination, functional recombination—are qualitatively different from the kind of reconceptualisation exemplified by the introduction of non-Euclidean geometry or special relativity, even granting that such innovations drew on prior materials [1, 34].

5.4 Implications

The results have direct implications for deploying LLMs in discovery-oriented tasks. Within Kuhnian normal science [1]—systematic exploration of hypothesis space within an established paradigm—LLMs offer substantial value through exhaustive combinatorial coverage and perfect recall. The success of AlphaFold in predicting protein structures [35] illustrates how AI can achieve transformative impact within a well-defined problem space, yet that system still required the conceptual primitives of physical chemistry and evolutionary co-evolutionary constraints that were established by prior human science. For paradigm-shifting innovation, however, the constrained-manifold pattern suggests that the required conceptual moves—recognising framework inadequacy, proposing new primitives, evaluating whether new frameworks are coherent—are not achievable by current architectures without external conceptual input.

Faggin’s framework [36] provides one theoretical lens. Faggin argues that genuine understanding requires phenomenological access: first-person awareness of what symbols mean, not only the ability to manipulate them via statistical regularities. This perspective is relevant here because it articulates why systems trained on distributional co-occurrence may face a principled barrier to ontological novelty: without semantic self-access, a model may be unable to detect that its current primitive ontology is insufficient. Whether this reflects the absence of consciousness in current LLMs or a narrower architectural limitation remains an open empirical question.

5.5 Planned Expert Validation Study

To address external validity of automated labels, future work will run a stratified expert annotation study on 30 proposals (5 sampled from each major structural

grouping). Three annotators are planned: one sensory biologist, one philosopher of perception, and one robotics researcher. Each proposal will receive a binary novelty judgment plus free-text justification. Inter-rater reliability will be quantified with Krippendorff’s α . Disagreements between expert labels and automated classification will be analyzed as directional bias (false-novel vs. false-traceable) and used to recalibrate thresholding and decomposition rules.

5.6 Limitations

Several methodological limitations constrain the conclusions.

Single domain. The study uses one conceptual domain (sensory modalities). Findings may not generalise to conceptual domains with different structural properties, such as mathematical objects, physical theories, or social organisational forms.

Automated evaluation. All evaluation methods are automated. Conservative thresholds mean we likely underestimate AI creativity, but the methods cannot capture all nuances an expert might perceive. Future work should include expert validation of a subset of proposals to calibrate and improve automated methods.

Corpus scope. Traceability is currently defined relative to academic mainstream literature in sensory biology, robotics sensing, and computational neuroscience. Non-academic sources (e.g., science-fiction corpora, patent databases, and broader philosophical text collections) were not included and may contain additional apparent predecessors. In addition, OpenAlex/Crossref coverage and metadata quality are heterogeneous across venues and years (including abstract availability and indexing lag), which can bias corpus composition and therefore observed traceability rates.

Prompt robustness. Full prompt-paraphrase grids were not included in the canonical experimental block. Conclusions should be read as structural tendencies under the present protocol, not as invariances proven over every prompting regime.

Temperature range. Temperature ablations were not systematically applied; runs were conducted at $T = 0.7$. Extending to broader temperature ranges and verifying that the null result persists would strengthen confidence.

Generative architecture coverage. All tested models are transformer-based autoregressive LLMs. The No New Basis Theorem applies to a wider class of gradient-descent learners, but empirical coverage of non-autoregressive or hybrid architectures is absent.

Interpretation of the No New Basis Theorem. The theorem provides formal motivation for the constrained-manifold interpretation but rests on assumptions (fixed vocabulary, continuous gradient-based learning) that may not hold for future architectures incorporating discrete search, external memory, or online learning components.

6 Conclusions

Current large language models demonstrate facility with lexical and structural creativity—recombining, extending, and remixing known conceptual elements in novel configurations. In this controlled sensory-modality task, none of 79 distinct normalized proposals achieved the study’s strict framework-external novelty classification.

Universal hull membership, frequent structural recombination, and non-saturated but substantial literature retrieval jointly support a constrained-manifold interpretation. The result is scoped to the tested ontology, prompt, corpus, and operational definitions; it should not be generalized into a universal prohibition on model creativity or future architectural development.

Declarations

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Competing Interests

The authors declare no competing interests.

Ethics Approval

Not applicable.

Author Contributions

All authors contributed to design, analysis, and writing.

Data and Code Availability

Data, prompts, model responses, and analysis code are publicly available at https://github.com/JABE22/PhD/tree/main/research/test1_ontological-innovation.

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